

# Shrouded Prices and Firm Reputation: Evidence from the U.S. Hotel Industry

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January 12, 2018

## Abstract

Firms in many industries engage in price obfuscation – tactics that intentionally make prices more difficult for consumers to discern. While existing research has focused on the short-term financial gains that motivate firms to obfuscate, reputational concerns may at least partially counteract these incentives if consumers punish deceptive firms via loss of loyalty in future transactions and/or publicly observable negative feedback. This paper addresses the latter possibility, exploring the impact of shrouded surcharges on firm reputation in the U.S. hotel industry. Using data collected from two major online travel sites, I exploit differences in surcharge disclosure across booking channels to identify the causal effect of hidden “resort fees” on traveler ratings. I find that hidden fees decrease ratings by roughly 0.15 points (on a rating scale ranging from 1 to 5). The magnitude of this effect varies based on firm characteristics, and this variation is consistent with observed heterogeneity in resort fee adoption patterns: when the expected punishment is more severe, firms are substantially less likely to adopt shrouded surcharges. Results shed light on the extent to which reputational mechanisms may act as a check against price obfuscation and other similar practices intended to exploit boundedly rational consumers.

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# 1 Introduction

A growing literature explores the various ways in which firms can (and do) utilize tactics that make prices more difficult, confusing, or time-consuming to discern. Fine-print investment fees cost unaware consumers tens of thousands of dollars over the course of their lifetimes. Retailers of consumer goods from mattresses to groceries strategically label products using inconsistent descriptors and units of measurement – making price comparisons challenging. And of course nearly anyone who has shopped online has experienced the frustration of having a large shipping charge or service fee tacked on just before check-out.<sup>1</sup>

Theory suggests that price obfuscation should be optimal in many settings (e.g., Ellison, 2005; Gabaix and Laibson, 2006; Spiegel, 2006; Carlin, 2009; Ellison and Wolitzky, 2012; Piccione and Spiegel, 2012; Chioveanu and Zhou, 2014).<sup>2</sup> Gabaix and Laibson’s work is probably the most relevant to this paper given its focus on shrouded surcharges, and a particularly striking result of their model is confined to a footnote: firms will *always* shroud unavoidable (i.e., mandatory) surcharges if there is even a single “myopic” consumer in the market to be exploited. Indeed, empirical evidence seems to suggest that shrouding mandatory surcharges is highly effective at boosting demand and markups (e.g., Morwitz, Greenleaf, and Johnson, 1998; Lee and Han, 2002; Xia and Monroe, 2004; Hossain and Morgan, 2006; Chetty et al., 2009;<sup>3</sup> Brown, Hossain, and Morgan, 2010; Kalayci and Potters, 2011).<sup>4</sup> In many instances, implementing a shrouded surcharge boosts seller revenue dollar-for-dollar. These empirical findings are in accordance with the Gabaix and Laibson conclusion that firms have an (initial) incentive to adopt shrouded pricing, but both bodies of literature seem disconnected from the empirical reality that firms in many settings do *not* universally or gratuitously adopt price obfuscation. Instead, the prevalence of shrouded surcharges varies widely, both within and across industries. Plainly put, some firms shroud, and some do not.

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<sup>1</sup>For a broader overview of various hidden surcharges across industries, see Greenleaf et al. (2015).

<sup>2</sup>Ellison (2006) and Grubb (2015) both provide fairly extensive surveys of key contributions in this literature.

<sup>3</sup>This is not a paper about obfuscation *per se*, but its results shed light on the effectiveness of shrouding mandatory components of price.

<sup>4</sup>See Ahmetoglu et al. (2014) for a review of this body of literature.

How might this be explained? Under what conditions and to what extent might market forces fully or partially counteract the incentives that motivate firms to obfuscate?

The literature on reputation and relational contracting has emphasized the way in which potential loss of future business can incentivize honesty in transactions (e.g., Telser, 1980; Klein and Leffler, 1981; MacLeod, 2007; Heyes and Kapur, 2012). Perhaps, then, reputational concerns may mitigate the motivation to obfuscate if consumers perceive these tactics as deceptive and subsequently punish the offending firms. Several contributions support the notion that consumer concerns about fairness and honesty are, indeed, highly relevant (e.g., Kahneman, Knetsch, and Thaler, 1986; Lee and Han, 2002; Xia and Monroe, 2004; Heo and Lee, 2011). In repeated transaction settings, embittered consumers might simply take their business elsewhere. Ian Ayres and Barry Nalebuff articulate this precise idea in their 2003 article, “In Praise of Honest Pricing.” A 2006 Boston Globe piece quotes Nalebuff reiterating a key message of this essay: “In the end, you don’t fool the customers with the hidden price . . . and if they feel ripped off, they won’t come back.”<sup>5</sup> Alternatively, some markets provide consumers with the ability to punish via publicly observable negative feedback (e.g., via rating mechanisms such as Yelp or TripAdvisor). Here, firms may have an incentive to avoid obfuscation even if interactions are not repeated and the set of buyers in the market is different each period.

This paper focuses on the latter set of circumstances and attempts to shed light on the potential reputational consequences of one specific type of price obfuscation – shrouded surcharges.<sup>6</sup> My empirical setting is the U.S. hotel industry, where the increasing prevalence of hidden fees has provoked substantial negative media attention and regulatory scrutiny in recent years. Of particular interest are “resort fees,” which are mandatory surcharges imposed by roughly 7% of hotels. These fees are typically shrouded in fine print, and both

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<sup>5</sup>Shea (2006). Laibson’s response in the same article: “Companies that hide fees may lose some customers, but along the way it’s a good ride.”

<sup>6</sup>The practice of obfuscating the total price of a good by breaking it into multiple (partially shrouded) components has been referred to in various literatures by a number of terms: drip pricing, partitioned pricing, add-on pricing, etc. In this paper, I most often refer simply to “hidden fees.”

news coverage and actual traveler review content indicates that many consumers are unaware of them at the time of booking. Instead, uninformed consumers find out about resort fees upon arrival at the hotel, or even, in some cases, upon check-out when receiving their final bill – anecdotally, at least, resulting in substantial indignation. To empirically quantify this “hidden fee effect” on firm reputation, I utilize online ratings (i.e., publicly observable scores that hotel travelers submit about their experience *ex post*) from two sources: Expedia and TripAdvisor.

I first document a robust negative relationship between resort fees and average traveler ratings in cross-sectional data. While this negative association persists even after controlling for observable firm and market characteristics, there is ample reason to be concerned that a firm’s decision to charge resort fees may be correlated with unobserved factors that affect ratings (e.g., quality of management) or that having a lower reputation is actually what causes these firms to adopt resort fees in the first place. To address these issues, I exploit differences in surcharge disclosure across booking channels. For a subset of hotels, Expedia’s booking platform actually includes resort fees in the price that travelers must pay up front – in other words, resort fees are unshrouded for customers booking these hotels through Expedia. And since travelers must have verifiably booked through an Expedia interface to submit an Expedia review, ratings on Expedia effectively serve as a control group for this hotel subset. In contrast, resort fees *are* hidden from consumers booking these same hotels via almost all other channels. Any associated “hidden fee effect” will accordingly be reflected in more negative reviews on sites such as TripAdvisor, where travelers from many different booking channels submit ratings.

This unique feature of the online hotel booking market allows me to estimate the causal effect of hidden fees in a difference-in-differences framework. By collecting repeated monthly cross-sections of hotel characteristics, I observe when a hotel adopts (or drops) a resort fee that is *not* hidden on Expedia but *is* hidden in other settings. Focusing on this key subset of hotels, I then compare time-stamped individual traveler reviews from Expedia (the control

group) to those from TripAdvisor (the treatment group) pre- and post-policy change. Both groups of reviews absorb any potential unobserved shocks in quality correlated with changes in resort fee policy, but only TripAdvisor reviews reflect the actual effect of hidden fees on ratings.

Results from this analysis indicate that hidden fees reduce traveler ratings by roughly 0.15 points (on a rating scale that ranges from 1 to 5). Applying benchmarks from existing literature (e.g., Luca, 2011; Anderson, 2012; Expedia, 2012), the revenue loss associated with this baseline decline in ratings is in the ballpark of 1-2% – a non-trivial amount, but still likely not enough in many cases to outweigh the revenue *gains* associated with the adoption of a hidden fee. The magnitude of the estimated effect on ratings varies substantially, however, depending on firm characteristics. In particular, lower-end hotels that adopt resort fees are punished much more severely than their higher-end counterparts. This finding helps begin to explain observed variation in resort fee adoption patterns within the broader market, where lower-end hotels rarely adopt these surcharges.

The remainder of this essay is structured as follows: In Section 2, I offer a brief overview of the workings of the U.S. hotel industry relevant to the present empirical analysis (including some background on Expedia and TripAdvisor – the two sources of data utilized here). Section 3 underscores broad cross-sectional evidence in support of the hypothesis that hidden fees result in lower ratings. The core empirical analysis is contained in Section 4; here, I introduce the difference-in-differences (D-in-D) model, present estimates for the causal effect of hidden fees on ratings, and discuss key empirical findings. Section 5 concludes.

## **2 Industry Background**

### **2.1 Hotel Fees: A Brief Overview**

Shrouded surcharges in the hotel industry are relatively common and increasing in prevalence. Total U.S. hotel revenue from fees (both mandatory surcharges and fees for

avoidable add-on goods – e.g., wireless internet, parking, late check-out, infant cribs, etc.) has doubled over the last decade or so, reaching \$2.45 billion in 2015. In particular, mandatory surcharges, most commonly dubbed “resort fees,”<sup>7</sup> have received a flurry of negative media attention. The American Hotel & Lodging Association reports that roughly 7% of hotels charge resort fees – a figure in line with what I observe in the data utilized here. Unlike room rates, which often fluctuate daily, resort fees are typically a flat dollar amount charged per night. Historically, these fees were charged only by genuine “resorts,” and were justified to customers as necessary to maintain expansive pool, spa, and fitness facilities. Today, however, many hotels that are *not* resorts in any traditional sense have adopted these surcharges. And while resort fees usually include access to Wi-Fi, other amenities such as parking, fitness facilities, pools, and spas may or may not be included depending on the individual property. Indeed, many properties charging resort fees today do not even *offer* such amenities, even for an additional charge.

Partitioning total price into a base price and a mandatory resort fee is attractive to hotels primarily because it allows them to advertise a low room rate up front. This low rate is the price that is visible to consumers during search. The additional fee, in contrast, is typically not disclosed until later in the booking process – often after the consumer has provided all necessary booking information, and even then, often only in fine print. Figure A1 in the Appendix outlines a timeline of the steps that take place in a typical transaction between customer and hotel. Because payment is generally not due at the time of booking but rather at the time of *departure* from the hotel, many buyers may not notice resort fee surcharges when making their purchase decision. And even if buyers are aware of the fees at the time of booking, it is likely that many do not fully account for them until payment is actually due (as in Morwitz, Greenleaf, and Johnson (1998), where consumers fail to fully account for partitioned surcharges). News coverage and anecdotal evidence from traveler reviews suggest

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<sup>7</sup>Resort fees are also sometimes known by other names such as “convenience fees,” “service fees,” “amenity fees,” “facilities fees,” or “destination fees.” Other mandatory surcharges include energy fees and cleaning fees; I focus only on resort fees in this paper, however, as they are by far the most common type of mandatory surcharge in the hotel setting.

that consumers are often, indeed, both surprised and angry about resort fees upon the receipt of their final bill.

## 2.2 Expedia and the OTA Channel

This paper utilizes data on U.S. hotels collected from a major online travel agency (OTA), Hotels.com (a subsidiary brand of Expedia).<sup>8</sup> In the U.S., 15-16% of individual consumer hotel bookings are conducted through OTAs,<sup>9</sup> which are web-based platforms that allow travelers to search for, compare, and reserve travel-related goods such as hotels, airfare, and rentals cars. In addition to Hotels.com, Expedia’s brands include Expedia.com, Hotwire, Travelocity, and Orbitz. As of year-end 2015, these platforms accounted for roughly 8% of total hotel bookings in the U.S. (equivalent to a 50-60% share of the U.S. OTA market), with Hotels.com representing the largest portion.<sup>10</sup>

Expedia and its subsidiary brands conduct the majority of their hotel bookings via a “pay now” merchant model: Expedia contracts with hotels to sell blocks of inventory, and travelers pay for these bookings *up front* directly through the OTA website, with Expedia as the merchant of record.<sup>11</sup> In contrast, major OTA competitors such as Booking.com primarily utilize an agency model in which travelers book rooms via the OTA but do not pay until they arrive at the hotel property – a process that more closely resembles the way in which the majority of bookings occur outside the OTA channel (where payment is typically not

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<sup>8</sup>Throughout the remainder of the text, I refer to the entities “Hotels.com” and “Expedia” interchangeably. As a practical manner, the informational content that is relevant for this analysis is equivalent across Hotels.com and Expedia.com (Expedia’s two largest hotel booking platforms). Most importantly, the presentation/handling of pricing and payments (including resort fees) is the same, and traveler reviews are pooled across the two sites (i.e., reviews from Expedia.com are included in the ratings/reviews listed on Hotels.com and are thus captured in my data). In this sense, the analysis presented here effectively applies to the segment of the market that books via either site.

<sup>9</sup>Source: TravelClick, 2015-2016 data.

<sup>10</sup>Source: Expedia Q4 2015 investor presentation (February 10, 2016).

<sup>11</sup>Prior to 2012, *all* of Expedia’s hotel bookings were conducted via the merchant model. In 2012, Expedia also introduced a “pay later” (agency model) option. As of 2015, an Expedia market manager reported that roughly 40% of U.S. hotels were enrolled in both the merchant model and the agency model, while roughly 60% continued to utilize only the merchant model. Of the 40% that provided both payment options, approximately 60% of bookings continued to flow through the “pay now” merchant model. This split suggests that roughly 80% of total hotel bookings on Expedia were conducted via the merchant model as of 2015.

due until check-out at the end of the hotel stay). This distinction will become important in Section 4, as it has implications for the way in which resort fees are presented to (and processed by) consumers on Expedia versus other booking channels.

From the perspective of the researcher, the Hotels.com website is an excellent source of data on U.S. hotels. The website includes listings for more than 50,000 individual properties,<sup>12</sup> reflecting upwards of 90% coverage of the roughly 53,000 hotels in the U.S.<sup>13</sup> For the majority of these hotels, Hotels.com provides data on average cumulative traveler ratings, room rates, mandatory fees, fees for optional add-ons (Wi-Fi, parking, etc.), and a wide range of hotel characteristics. Monthly cross-sectional snapshots of this data captured between April 2015 and March 2016 allow me to observe variation in hotel resort fee policies (and other characteristics) over time.

In addition to the information described above, Hotels.com also provides reviews from individual travelers for most hotel properties. Each review contains the date of the traveler's stay, the date that the review was written, a rating between 1 and 5, (optional) written commentary from the reviewer, and, in some cases, information about the traveler's reason for travel. Importantly, to write a review on an Expedia-brand site, a traveler *must have verifiably booked through the Expedia platform*. This feature of the data (in combination with the fact that resort fees are handled differently on Expedia versus other booking channels for at least a subset of hotels) will facilitate identification in the difference-in-differences analysis presented in Section 4.

## 2.3 TripAdvisor

In Section 4's difference-in-differences analysis, I also utilize individual traveler review data from TripAdvisor. Unlike Hotels.com, TripAdvisor is not an OTA in the traditional

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<sup>12</sup>This figure reflects coverage as of March 2016. Hotels.com coverage was less expansive at the beginning of my data collection efforts in April 2015, at approximately 45,000 U.S. hotels.

<sup>13</sup>Source: American Hotel and Lodging Association, 2015.



sense.<sup>14</sup> Instead, it is a web-based aggregator of ratings for travel-related goods – hotels, attractions, and even restaurants. Travelers use TripAdvisor primarily to conduct research when planning trips and to rate their experiences afterwards. The site is massive, providing reviews for nearly one million hotels worldwide.<sup>15</sup>

The basic information contained in each individual TripAdvisor review largely mirrors that described for Expedia reviews above. A critical difference, however, is the ability for *anyone* to write a review on TripAdvisor. In empirical analyses, I thus use TripAdvisor reviewers as a proxy for non-Expedia purchasers. It is important to note, however, that there are, presumably, some Expedia buyers who write reviews on TripAdvisor, since the site is open to anyone. While the ideal dataset would guarantee mutual exclusivity between these two groups, the overlap here is not likely to be problematic in practice for two reasons. Firstly, having some control group observations (reviews from Expedia buyers) mixed in with the treatment group (reviews from non-Expedia buyers, proxied by TripAdvisor reviews) has the effect of biasing the estimated treatment effect towards zero in the difference-in-differences model specified here. So if anything, the treatment effect I estimate will be conservative in magnitude as a result of this issue. Secondly, the proportion of TripAdvisor reviewers that are Expedia buyers is likely to be relatively small, as Expedia brands account for only 8% of total U.S. bookings. Moreover, Expedia buyers receive email prompts to review their stay *on Expedia*, so they are arguably more likely to do so than they are to write a review on TripAdvisor. In practice, the percent of TripAdvisor reviews in my data that explicitly mention “Expedia” or “Hotels.com” in their review text is 0.5%.

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<sup>14</sup>TripAdvisor has begun to move in this direction, adding an “instant booking” feature in 2014. To date, however, this line of business constitutes only a small portion of TripAdvisor’s revenue, which comes largely from advertisements.

<sup>15</sup>Per TripAdvisor 2015 10-K filings.

### 3 Resort Fees and Ratings on Expedia: A Cross-Sectional Snapshot

While the crux of this paper’s empirical analysis focuses on what happens to traveler ratings at hotels where the resort fee policy *changes* over time, it is useful to begin by first examining a static cross-section of the overall U.S. hotel market for context.<sup>16</sup> Figure 1 illustrates a strong negative relationship between resort fees and cumulative average traveler ratings: of all U.S. hotels listed on Expedia, those that charge resort fees have lower average ratings in each major star category.<sup>17</sup>

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Insert Figure 1 About Here

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Of course there is ample reason for skepticism when interpreting these distributions – the decision to adopt resort fees is hardly exogenous. Hotels that adopt these fees tend to be higher-quality, higher-priced, independent (i.e., not chain-affiliated), larger properties. They also tend to be clustered in certain kinds of geographic markets (e.g., resort fees are particularly common in beach areas, in Las Vegas, etc.).<sup>18</sup> A natural question is whether or not the negative correlations depicted in Figure 1 hold up after controlling for these sorts of factors. Before moving forward, however, it is important to distinguish between the two distinct ways in which resort fees may be implemented on Expedia.

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Insert Figure 2 About Here

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<sup>16</sup>All figures, summary statistics, and cross-sectional analyses presented here are based on a cross-section of data collected in July 2015. Any of the 12 cross-sections collected between April 2015 and March 2016, however, yield substantively similar empirical findings.

<sup>17</sup>See Table A1 for more information on star categorizations. 97% of hotels in the sample fall between 2 and 4.5 stars and are thus captured in one of the three major star categories illustrated. For the remaining 1,531 hotels at the very low end (1-star and 1.5-star) and very high end (5-star), a t-test does not reject the null hypothesis that the means for the resort fee vs. non-resort fee groups are the same.

<sup>18</sup>Chiles (2016) addresses the competitive market forces that potentially influence a firm’s incentives to obfuscate. These issues are beyond the scope of this paper, and I abstract away from them with the use of market-level fixed effects in regression specifications.

Recall from Section 2 that most customers who purchase hotel rooms on Expedia actually pay at the time of booking versus at the hotel. As such, hotels that charge a resort fee may choose to have customers pay Expedia the fee up front along with the room rate (the fee is “included” or “unshrouded”), or, alternatively, to wait until the customer arrives at the property to charge them the fee directly (the fee is “separate” or “shrouded”). Figure 2 provides a comparison of the screens that consumers encounter in each of these cases.<sup>19</sup> As illustrated, “separate” fees are shrouded in fine print on Expedia at the time of booking, and since customers are not asked to pay them up front, they are easy to miss. In contrast, “included” fees are, at least for buyers on Expedia, *not hidden* at the time of booking, since the total price (inclusive of fees) is both presented and charged up front. This practice – actually requiring payment of the resort fee at the time of booking – effectively unshrouds the fee and is almost entirely unique to the Expedia booking channel. I thus define two variables to capture these two distinct implementations: *resort\_fee\_sep* (for when the fee is “separate,” i.e., shrouded) and *resort\_fee\_inc* (for when the fee is “included,” i.e., not shrouded).<sup>20</sup>

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Insert Table 1 About Here

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Insert Table 2 About Here

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Tables 1 and 2 present summary statistics and correlations for both resort fee variables as well as other key firm-level variables derived from Expedia data (see Table A1 in the

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<sup>19</sup>For further context, Figure A2 in the Appendix provides an illustration of these respective transactional timelines. To understand the incentives a hotel faces when determining which resort fee payment structure to select for its Expedia bookings, recall that Expedia makes money by taking a percentage of the booking value. Thus, on the one hand, the incentive for a hotel *not* to include the resort fee in the amount paid up front is obvious: firms are not charged a commission on the fee if it is collected at the property versus through the Expedia interface. On the other hand, trying to collect the fee once a guest arrives may be more of a hassle for firms, especially if guests are, indeed, irritated about the additional charges. Ultimately this decision is at the discretion of the revenue manager at each individual hotel property, and, like the decision to charge a resort fee in the first place, it may be correlated with other factors that affect ratings.

<sup>20</sup>Note that if a hotel does not charge a resort fee at all, both *resort\_fee\_sep* and *resort\_fee\_inc* are equal to 0.

Appendix for detailed variable descriptions). Notably, the unconditional correlation between ratings and resort fees (of both varieties) is actually positive, due to the fact that the adoption of resort fees is highly associated with more fundamental drivers of ratings like star category (i.e., a rough proxy for “quality”) – hotels in higher-tier star categories are substantially more likely to utilize resort fees. Taking into account the need to control for these sorts of factors, Table 3 presents results for various specifications of the following model, where  $X_j$  is a vector of firm and market characteristics corresponding to hotel  $j$ :

$$avg\_rating_j = \beta_0 + \beta_1(resort\_fee\_sep_j) + \beta_2(resort\_fee\_inc_j) + \alpha'X_j + \varepsilon_j \quad (1)$$

In addition to the key variables defined in Table A1, I include a much more detailed set of firm controls in some specifications (defined in Table A2). In several specifications, I also include fixed effects for chain affiliation; more details about this can be found in Table A3.

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Insert Table 3 About Here

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In Table 3, the coefficient on *resort\_fee\_sep* turns negative after controls for “first-order” determinants of ratings are added in Model (2) (star category, number of rooms, and chain affiliation, which together account for roughly 56% of the variation in average ratings across hotels). As additional controls at both the firm-level and the market-level are added, the magnitude of the estimate on *resort\_fee\_sep* actually increases slightly (to roughly -0.05 points); its statistical significance also increases markedly with the introduction of these additional controls.<sup>21</sup> In contrast, the coefficient on *resort\_fee\_inc* is insignificant (or, in some specifications, only marginally significant). This is in line with the narrative that there is a “hidden fee effect” driving the negative relationship between ratings and fees rather than some unobserved factor. When resort fees are shrouded, they are associated with a statistically significant decline in ratings of 0.05 points. When resort fees are *not* shrouded, ratings do

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<sup>21</sup>Note that in all cases throughout this paper, *price* and *log-price* reflect base prices – i.e., the price before resort fees are added on.

not fall by any statistically significant amount.

## 4 Hidden Fees and Ratings: Estimating the Relationship

While the cross-sectional evidence is clearly suggestive, it is not sufficient to conclude that there is a causal relationship between hidden fees and firm reputation. For one, while the introduction of controls does not seem to effect the estimates in any substantial way, one factor that I have no means of controlling for is quality of management. If, for example, we believe that “bad” managers might be more likely to implement resort fees and also more likely to offer hotel experiences that are worse on unobservable dimensions, then this would confound results in the cross-sectional analysis. Moreover, in cross-sectional data, it is impossible to ascertain the order of events: perhaps lower ratings drive hotels to adopt resort fees instead of the posited opposite relationship. Thus, to identify the true causal effect of shrouded fees on ratings, I focus on trends in individual traveler ratings at hotel properties where resort fee policies *change*. As noted in Section 2, repeated cross-sectional datasets captured monthly between April 2015 and March 2016 allow me to observe variation in hotel resort fee policies (and other characteristics) over time. Table 4 provides a summary of the changes in resort fee policy that occurred over this period.

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Insert Table 4 About Here

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In line with news coverage, the prevalence of resort fees is, on net, increasing: 286 U.S. hotels adopted resort fees during the sample window, while 133 discontinued the use of these surcharges.<sup>22</sup> For the 419 hotels that either added or dropped resort fees, I merge the panel

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<sup>22</sup>In addition to hotels that adopt resort fees and drop resort fees, there are also some hotels that appear to make multiple changes over the course of the sample window (e.g., adopt a fee, then delete it, then adopt it again). This is not very common, however, and I do not include these hotels in the analysis here (partly out of concerns for the integrity of the data when unusual changes like this appear to occur, and partly out of a desire to keep the empirical analysis straightforward and intuitive for readers. The inclusion of these hotels in the analysis does not substantively change any results.

of information on hotel characteristics described above with individual traveler review data from both Expedia and TripAdvisor. Since information on the date of the traveler’s stay is provided in the review data, I can deduce whether or not resort fees were in effect at the time an individual reviewer stayed at the hotel property in question.

For each hotel, I focus on reviews from the twelve months leading up to the policy change plus those from the three months following the policy change. The set of 419 hotels corresponds to a sample of 120,498 individual traveler reviews (an average of 288 reviews per hotel).<sup>23</sup> This review-level data contains each individual traveler’s rating (importantly, distinct from the *average cumulative* rating metric utilized in the cross-sectional analysis) as well as the nature of the traveler’s trip (business, couple, family, friends, or other/unspecified). Table 5 below provides summary statistics for key review-level variables, and Table A4 in the Appendix provides precise variable definitions.

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Insert Table 5 About Here

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The preceding section discussed resort fee implementation on Expedia. For the purposes of the subsequent analysis, it is also important to establish how resort fees are presented *outside* the Expedia setting. In the case of both *resort\_fee\_sep=1* and *resort\_fee\_inc=1*, resort fees are unlikely to be disclosed clearly for customers booking via channels other than Expedia. This is important for identification, as it allows me to exploit variation in resort fee disclosure when these fees are *not* shrouded on Expedia. Figure 3 illustrates an example of this disparity: even when resort fees are “included” (i.e., not shrouded) on Expedia, they are typically hidden in other booking environments.

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Insert Figure 3 About Here

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<sup>23</sup>65,040 from Expedia and 55,458 from TripAdvisor.

## 4.1 A Difference-in-Differences Model

The left portion of Figure 4 illustrates the intuition behind the difference-in-differences framework utilized here.<sup>24</sup> In this scenario, resort fees are *not actually hidden* from Expedia reviewers, allowing these reviewers to effectively function as a control group. In contrast, travelers booking via non-Expedia channels *are* subject to the hidden fee “treatment,” and any subsequent effect is reflected in lower ratings on TripAdvisor. This allows us to identify the effect of hidden fees on ratings: both sets of reviews absorb any potentially confounding factors associated with resort fee adoption, but only TripAdvisor reviews reflect the actual hidden fee effect. And, indeed, in the left portion of Figure 4, we see ratings fall post-adoption at TripAdvisor, but actually edge up a bit at Expedia.

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Insert Figure 4 About Here

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In the right portion of Figure 4, we see ratings over time at both Expedia and TripAdvisor for those hotels that adopted “separate” resort fees. In this case, when a hotel adopts a fee, we observe a negative effect on ratings in both groups of reviewers. This is in line our existing intuition. If the fee is “separate,” then all travelers pay it at the hotel – it is shrouded from both Expedia reviewers and TripAdvisor reviewers at the time of booking, and both groups appear to punish firms via lower average ratings following adoption. This pattern is consistent with cross-sectional results, but we cannot actually identify the effect of hidden fees on ratings in this scenario. Both groups of reviewers receive the hidden fee “treatment,” so it is impossible to separate out the impact of hidden fees from any other potential confounding factors that may be correlated with their adoption.

I thus focus my analysis on the sub-sample of hotels<sup>25</sup> that make policy changes involving “included” resort fees, as it is only within this subset that we can actually identify the effect

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<sup>24</sup>Note that these charts focus only on hotels that *adopt* fees for the sake of illustration purposes; the empirical analysis includes both hotels that adopt fees and hotels that drop fees.

<sup>25</sup>171 in total: 104 that adopted (illustrated in Figure 4) and 67 that dropped.

of hidden fees on ratings. Within this sample, I estimate the following model for traveler  $i$ 's rating of firm  $j$  in period  $t$ :

$$rating_{ijt} = \beta_0 + \beta_1(resort\_fee_{jt}) + \beta_2(tripadv_{ijt}) + \beta_3(resort\_fee_{jt} * tripadv_{ijt}) + \varepsilon_{ijt} \quad (2)$$

In the above specification,  $\beta_1$  estimates the main effect of resort fees on ratings; this term absorbs fluctuations due to confounding factors that may be correlated with resort fee adoption and will be positive if hotel quality increases (on unobserved dimensions) in conjunction with the adoption of fees and negative if quality decreases.  $\beta_2$  accounts for systematic differences in ratings from Expedia reviewers versus TripAdvisor reviewers (i.e., differences in the control group and treatment group). And  $\beta_3$  is our treatment effect, measuring the causal effect of hidden fees on ratings. To increase the model's precision, I also include fixed effects and control variables in various specifications.

## 4.2 Key Empirical Results

Table 6 provides results for multiple specifications of the D-in-D model detailed above. A negative coefficient on the interaction term  $tripadv * resort\_fee$  – the “hidden fee” treatment – is in line with the notion that shrouded prices harm ratings. The adoption of a resort fee is associated with a fall in subsequent traveler ratings of roughly 0.15 points (or, conversely, the discontinuation of a resort fee is associated with an increase in ratings of this same magnitude). In practical terms, this means that TripAdvisor ratings fall roughly 0.15 points *more* than Expedia ratings when a hotel adopts a hidden resort fee of the “included” variety. This effect remains quite stable (and significant) as various controls are added to the model. (E.g., the introduction of firm fixed effects in Model (2) dramatically increases R-squared (from .01 to .21), but the coefficient estimate for our treatment effect is largely unchanged.)

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Insert Table 6 About Here

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One additional result here is that the main effect on *resort\_fee* is actually positive, albeit marginally so (0.05 points). In other words, absent the “hidden fee effect,” ratings actually rise a bit in conjunction with the adoption of a resort fee. How should this finding be interpreted? A comparison of hotel characteristics pre- vs. post-policy change helps to shed light on this issue. While hotel traits such as chain affiliation and amenities do not tend to change much over time at the vast majority of properties, there are some instances of churn. Table 7 details differences in average firm characteristics when resort fees are present versus when they are not.<sup>26</sup> Here we see that in almost every case, the adoption of resort fees is correlated with the provision of *more* amenities. In particular, resort fees seem to be correlated with the adoption of (relatively) easy-to-implement amenities such as concierge services and beach/pool amenities (e.g., cabanas, towel service, etc.). Resort fees are also marginally correlated with the provision of more capital-intensive amenities (e.g., pools, tennis/golf facilities), but these types of changes over time are substantially less common. In any case, all else equal, we would expect the provision of more amenities to contribute to higher ratings on average. And while all of the characteristics detailed in Table 7 are observed and incorporated into Models (4) and (5), if hotels are also increasing quality on *unobserved* dimensions, then this may help to explain the positive coefficient estimate for *resort\_fee*.

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Insert Table 7 About Here

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An implicit assumption of the D-in-D identification strategy utilized here is that the change in the control group post-treatment is an appropriate proxy for the counterfactual change that *would* have occurred in the treatment group had there been no treatment. Several arguments can be made for why this assumption seems reasonable in this setting. For one, ratings from Expedia and TripAdvisor appear to exhibit parallel (and roughly flat) trends prior to the resort fee “treatment” in the D-in-D sample (left half of Figure 4). To further

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<sup>26</sup>For the sake of clarity in diction, I describe these changes from the point of view of a hotel adopting a resort fee, but the reverse will be true for hotels dropping resort fees.

examine the robustness of this observation, I estimate a model (for the subset of hotels that adopt fees) where *tripadv* (i.e., the indicator for whether or not a review comes from the treatment group) is interacted with a set of time-to-treatment dummy variables. As depicted in Figure 5, coefficient estimates on this set of interaction terms are not statistically different from zero prior to the policy change but become negative following resort fee adoption.

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Insert Figure 5 About Here

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In addition, reviewers on TripAdvisor and Expedia appear to be similar in terms of observable characteristics. Table 5 (introduced previously) includes a breakdown of trip types for Expedia and TripAdvisor reviewers, which are not statistically different within the set of reviews where this information is specified. The distribution of ratings on the two sites, however, is markedly different. In particular, reviewers on TripAdvisor are much less likely to assign a middling review of two, three, or four, and much more likely to award a five. This is potentially explained by the fact that TripAdvisor and Expedia likely have different frequencies of “fake” reviews – deceptive reviews posted by businesses in an effort either to inflate their own average ratings or damage those of their competitors. As Mayzlin, Dover, and Chevalier (2014) point out, the cost of writing a fake review is substantially lower on TripAdvisor than on Expedia, as the latter requires actually paying to book a room through an Expedia interface. This is not necessarily problematic to the D-in-D framework, so long as the propensity for fake reviews on either site does not change in a manner that is correlated with changes in resort fee policies. To test for this empirically, I randomly select 150 reviews from each of four categories (Expedia pre- and post-policy change and TripAdvisor pre- and post-policy change) and manually run them through *reviewskeptical.com*, a website designed by Cornell University computer science researchers to test for fake hotel reviews (with claims of 90% accuracy). Table 8 details results. As expected, TripAdvisor seems to have more deceptive reviews than Expedia, but importantly, the frequency of fake reviews does not

appear to change with the adoption of a resort fee.

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Insert Table 8 About Here

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Perhaps the most compelling argument in defense of the assumptions inherent to the D-in-D specification is the right half of Figure 4 (in which both Expedia and TripAdvisor reviewers are “treated” with the hidden fee), which depicts the two reviewer groups reacting similarly when exposed to the same shock. While it is not possible to formally identify a treatment effect in this case since both groups the hidden fee “treatment,” it is still useful to quantify the correlation between ratings and “separate” fees as a point of reference. Running Model (5) as specified above on the set of reviews for hotels that made policy changes involving a “separate” resort fee, the coefficient on the interaction term (*tripadv \* resort\_fee*) is insignificant (in other words, ratings continue to move in parallel - exactly what we would expect). The coefficient estimate on the main effect, *resort\_fee*, is -0.09 (with a t-statistic of -4.5 after adjusting standard errors for 237 hotel-clusters). This is also very much in line with what we would expect to observe absent the D-in-D framework, where we estimate in Model (5) a main effect of 0.05 and a treatment effect of -0.13 – summing to a total observed effect of -0.08. Indeed, all of these results also tie closely to the results obtained in the cross-sectional analysis of Section 3, where we estimated a coefficient for resort fees of -0.05 within the aggregate U.S. market. This consistency between all three sets of results lends confidence to the results in Table 6.

### 4.3 Treatment Effect Heterogeneity and Potential Mechanisms

If one of the objectives of this paper is to better understand why some firms adopt shrouded pricing and some don’t, it is essential to explore how this estimated treatment effect may vary based on firm characteristics. To do this, I apply the D-in-D specification in Model (5) above to several key sub-samples of the data. Table 9 presents results for this analysis.

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Insert Table 9 About Here

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The top section of Table 9 shows that the punishment in ratings is roughly three times more severe for lower-tier hotels (-0.3 points) versus mid- and high-tier hotels (roughly -0.1 points). These results are consistent with heterogeneity in resort fee adoption patterns in the broader market – in particular, hotels in low star categories are substantially less likely to adopt resort fees. There are several plausible (and not mutually exclusive) explanations for why this might be. Lower-tier hotels tend to be less expensive; perhaps more price-sensitive customers react more unfavorably to being charged additional hidden fees. In addition, lower-tier hotels are substantially less likely to offer traditional “resort” amenities – health clubs, spas, golf courses, etc., so consumers may view resort fee surcharges as less fair in these settings and punish firms more severely.

While this empirical setup is not ideally suited to precisely pin down mechanism (as price and the provision of amenities are inherently correlated), the remaining sections of Table 9 attempt to shed some light. In the low-tier segment, lower-priced hotels<sup>27</sup> and higher-priced hotels receive similar punishments. Treatment effect estimates are also similar for hotels regardless of whether or not they provide resort amenities (however, only 3 hotels in this segment offer such amenities, so it is somewhat difficult to draw meaningful conclusions here). In the mid/high-tier segment, price again seems to be unimportant as a determinant of effect size. The provision of traditional resort amenities does seem to matter, though. In this market segment, hotels that do not provide these amenities account for the majority of the observed negative relationship between resort fees and ratings, suggesting that a mechanism driven by concerns over fairness (e.g., similar to that proposed by Kahneman, Knetsch, and Thaler, 1986) may be driving results.

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<sup>27</sup>Relative to other hotels in their respective markets of the same star category.

## 4.4 Discussion

While the empirical evidence does indicate that shrouded surcharges result in lower ratings, the magnitude of this effect seems rather modest at roughly 0.15 points (on a rating scale that ranges from 1 to 5). Average ratings are just shy of 4.0, so the estimated treatment effect represents a decrease of 3-4% off of this mean value. It is important to note, however, that this estimated treatment effect is likely biased somewhat towards zero for two key reasons. The first was discussed in Section 2: some members of the treatment group (i.e., TripAdvisor reviewers) don't actually receive the hidden fee "treatment." This is because *anyone* can write a review on TripAdvisor, so some Expedia buyers (i.e., members of the control group) undoubtedly account for some portion of the reviews in this setting. There is also a strong argument that results are biased towards zero due to selection effects. Hotels for which punishment is likely to be large (e.g., lower-tier hotels, as illustrated in the previous sub-section) are substantially less likely to adopt fees. This is true both intuitively and empirically to the extent that we can observe. Turning this on its head implies that the aggregate treatment effect estimated here is smaller in magnitude than would be expected if resort fees could be randomly assigned to hotels.

If we suppose, however, that this bias is small, and -0.15 is a fairly accurate estimate of the true effect of hidden fees on ratings, then what are the implications for firm performance? In online settings in particular, seller reputation in the form of ratings has been shown to have a substantial impact on demand.<sup>28</sup> Several authors have studied this in the context of eBay, where there have been at least nine papers since 2001 (see Cabral and Hortacsu (2010) for details) documenting a positive correlation between seller reputation and sale prices, number of bids, and/or probability of sale in cross-sectional data. Outside the eBay setting, Luca (2011) uses a regression discontinuity design to study the relationship between Yelp ratings and restaurant revenue, concluding that a one-star increase on Yelp results in a 5-9% increase

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<sup>28</sup>See Dellarocas (2003) or Josang, Ismail, and Boyd (2006) for a more comprehensive review of the literature on online feedback mechanisms.

in sales. And in the hotel industry, the setting for this paper, ratings are known to play a critical role: using transactional data from an online travel agency, Anderson (2012) finds that if a hotel increases its review score by 1 point (on a 5-point scale), then it can increase its price by 11.2% and still maintain the same occupancy. Similarly, Expedia (2012) reports that a 1-point increase in review scores equates to a 9% increase in average daily room rates.

These studies collectively suggest that a 1-point change in ratings corresponds to roughly a 10% change in revenue. Scaling this figure linearly, if ratings decrease by 0.15 points upon adoption of a resort fee, then the associated decrease in revenue should be in the ballpark of 1.5%. How does this compare with potential *gains* in revenue due to hidden fees? The data here indicates that base room rate pricing effectively doesn't change when a fee is adopted (\$132 on average vs. \$135), and resort fees are, on average, roughly 10% of room rate. So if firms charge the same base prices and continue to attract the same number of customers (as prior literature indicates is likely), then the adoption of a hidden fee will directly boost revenue by 10%. It may be more realistic to assume that fees do dampen demand at least *slightly*,<sup>29</sup> so perhaps the increase in revenues associated with hidden fees is less than 10% after accounting for some dip in quantity. Yet even if this is the case, the financial benefits of obfuscation likely outweigh the reputational costs under these baseline assumptions – at least for many firms.<sup>30</sup>

Finally, it is important to address external validity: to what extent might these results apply in other industry settings and to other types of shrouding? Certainly these findings suggest that when online feedback mechanisms exist, consumers may be able to at least partially deter firms from engaging in practices that they deem to be undesirable by punishing them with lower ratings. For this to be the possible, though, consumers must, at the very least, *recognize* that they are being exploited. While this is likely the case in many contexts

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<sup>29</sup>Arguably some consumers do recognize the fee as an effective price increase and adjust their purchase decisions accordingly.

<sup>30</sup>Given the nature of the sample utilized here, this is not entirely surprising, as my results are derived only from data where firms have, at some point, chosen to adopt a resort fee (so ostensibly, the benefits *should* outweigh the costs for these firms). It still seems surprising, however, that only 7% of hotels in the U.S. have adopted resort fees in light of what seems to be a fairly lopsided cost-benefit tradeoff.

(particularly with regard to shrouded surcharges – ultimately consumers have to pay them, so in most cases they do not remain hidden forever), it may not hold in some of the most high-stakes settings. For example, in the case of hidden 401(k) fees, consumers may never actually realize that the fees exist if they are automatically deducted and individuals do not pay close attention to their account statements over time. This specific example also raises the point that there are many goods and services where online feedback mechanisms are not commonly utilized (e.g., financial services). Without this particular market feature, it is less clear what sort of punishment firms might face for behaving badly. Loss of loyalty in repeated transaction settings may be an effective deterrent, but this is not directly examined here.

## 5 Conclusions

The notion that consumers may punish firms for price obfuscation (and deceptive practices more generally) is hardly new, yet surprisingly little empirical research exists to support it. This paper provides evidence on one mechanism by which firms may be punished for deceptive pricing practices: reputational damage in the form of lower ratings. Empirical results indicate that the adoption of a hidden fee decreases subsequent traveler ratings by roughly 0.15 points (on a rating scale that ranges from 1 to 5). This effect is substantially larger for lower-tier firms, which helps to explain heterogeneity in resort fee adoption patterns (lower-tier firms are much less likely to adopt shrouded surcharges). Further sub-sample analysis suggests that the mechanism driving this punishment effect has to do with consumer concerns about fairness, of the kind described in Kahneman, Knetsch, and Thaler (1986) – i.e., consumers punish firms for shrouded surcharges more when these surcharges are perceived as less “fair.”

Benchmarks from existing literature suggest that a reduction in ratings by the baseline amount of 0.15 points is associated with a decrease in revenue on the order of magnitude of 1-2% – not an inconsequential figure, but still probably not enough to outweigh the revenue gains from price obfuscation (which are likely more in the ballpark of 5-10%). There are several potential interpretations for why so few hotels (only 7% in the U.S.) charge resort

fees despite the fact that the benefits appear to outweigh the costs so markedly. One is that the observed treatment effect here is, arguably, best understood as a lower bound for the true effect of hidden fees on ratings. Intuitively, all else equal, firms should be more likely to adopt hidden fees when the expected reputational damage is *lower*. Yet by definition, my relevant sample can only consist of firms that have, at some point, adopted a resort fee. Ostensibly, firms that expect punishment to be more severe never adopt fees at all, thus biasing the results estimated here towards zero.

In addition, there may be other important costs to obfuscation (e.g., loss of repeat business, logistical/labor costs associated with handling disgruntled patrons) besides the reputational costs explored here that might help explain why the resort fee adoption rate remains so low. Or perhaps many firms in the hotel industry are making a mistake and would actually be better off adopting shrouded surcharges. Indeed, it is possible that these markets are slowly converging towards new equilibria where many more firms *will* utilize shrouded pricing. A clear understanding of the extent to which market forces may (or may not) mitigate these outcomes is of fundamental importance to policymakers. Certainly more research is needed to further our understanding of these issues.



## References

- Ahmetoglu, G., A. Furnham, and P. Fagan. 2014. Pricing practices: A critical review of their effects on consumer perceptions and behaviour. *Journal of Retailing and Consumer Services*, 21(5): 696-707.
- Akerlof, G. and R. Shiller. 2015. *Phishing for fools: the economics of manipulation and deception*. Princeton University Press.
- Anderson, C. 2012. The impact of social media on lodging performance. *Cornell Hospitality Report*, 12(15): 6-11.
- Ayres, I. and B. Nalebuff. 2003. In praise of honest pricing. *MIT Sloan Management Review*, 45(1): 23-28.
- Brown, J., T. Hossain, and J. Morgan. 2010. Shrouded attributes and information suppression: evidence from the field. *The Quarterly Journal of Economics*, 125(2): 859-876.
- Cabral, L., and A. Hortacsu. 2010. The dynamics of seller reputation: evidence from eBay. *The Journal of Industrial Economics*, 58.(1): 54-78.
- Carlin, B. 2009. Strategic price complexity in retail financial markets. *Journal of Financial Economics*, 91(3): 278-287.
- Chetty, R, A. Looney, and K Kroft. 2009. Salience and Taxation: Theory and Evidence. *The American Economic Review*, 99(4): 1145-1177.
- Chiles, B. 2016. Resorting to obfuscation: why do firms adopt shrouded pricing? *Working Paper, available by request*.
- Chioveanu, I., and J. Zhou. 2013. Price competition with consumer confusion. *Management Science*, 59(11): 2450-2469.
- Dellarocas, C. 2003. The digitization of word of mouth: promise and challenges of online feedback mechanisms. *Management Science*, 49(10): 1407-1424.
- Ellison, G. 2005. A model of add-on pricing. *Quarterly Journal of Economics*, 120(2): 585-637.
- Ellison, G. 2006. Bounded rationality in industrial organization. *Econometric Society Monographs*, 42: 142.
- Ellison, G., and S.F. Ellison. 2009. Search, obfuscation, and price elasticities on the internet. *Econometrica*, 77(2): 427-452.
- Ellison, G., and A. Wolitzky. 2012. A search cost model of obfuscation. *The RAND Journal of Economics*, 43(3): 417-441.
- Gabaix, X., and D. Laibson. 2006. Shrouded attributes, consumer myopia, and information suppression in competitive markets. *Quarterly Journal of Economics*, 121(2): 505-540.

- Greenleaf, E., E. Johnson, V. Morwitz, and E. Shalev. 2015. The price does not include additional taxes, fees, and surcharges: A review of research on partitioned pricing. *Journal of Consumer Psychology*, 1(26): 105-124.
- Grubb, M. 2015. Failing to choose the best price: Theory, evidence, and policy. *Review of Industrial Organization*, 47(3): 303-340.
- Heo, C. and S. Lee. 2011. Influences of consumer characteristics on fairness perceptions of revenue management pricing in the hotel industry. *International Journal of Hospitality Management*, 30: 243-251.
- Heyes, A. and S. Kapur. 2012. Angry customers, e-word-of-mouth and incentives for quality provision. *Journal of Economic Behavior & Organization*, 84: 813-828.
- Hossain, T., and J. Morgan. 2006. Plus shipping and handling: revenue (non) equivalence in field experiments on eBay. *Advances in Economic Analysis & Policy*, 5.2.
- Josang, A., R. Ismail, and C. Boyd. 2006. A survey of trust and reputation systems for online service provision *Decision Support Systems*. 43: 618-644.
- Kahneman, D., J. Knetsch, and R. Thaler. 1986. Fairness as a constraint on profit seeking: Entitlements in the market. *The American Economic Review*: 728-741.
- Kalayci, K. and J. Potters. 2011. Buyer confusion and market prices. *International Journal of Industrial Organization*. 29: 14-22.
- Klein, B. and K. Leffler 1981. The role of market forces in assuring contractual performance. *Journal of Political Economy*. 89(4): 615-641.
- Lee, Y., and C. Han. 2002. Partitioned pricing in advertising: Effects on brand and retailer attitudes. *Marketing Letters*, 13(1): 27-40.
- Luca, M. Reviews, reputation, and revenue: The case of Yelp. com. (September 16, 2011). Harvard Business School NOM Unit Working Paper 12-016 (2011).
- MacLeod, B. 2007. Reputations, relationships, and contract enforcement. *Journal of Economic Literature*, 45(3): 595-628.
- Mayzlin, D., Y. Dover, and J. Chevalier. 2014. Promotional reviews: An empirical investigation of online review manipulation. *The American Economic Review*, 104.8: 2421-2455.
- Morwitz, V., E. Greenleaf, and E. Johnson. 1998. Divide and prosper: Consumers' reaction to partitioned prices. *Journal of Marketing Research*, 35: 453-463.
- Muir, D., K. Seim, and M.A. Vitorino. 2013. Drip pricing when consumers have limited foresight: evidence from driving school fees. Available at SSRN 2220986.
- Ott, M., Y. Choi, C. Cardie, and J. Hancock. 2011. Finding deceptive opinion spam by any stretch of the imagination. *Proceedings of the 49th Annual Meeting of the Association for Computational Linguistics*, 309-319.

Piccione, M. and R. Spiegler. 2012. Price competition under limited comparability. *Quarterly Journal of Economics*, 127(1): 97-135.

Resnick, P., Zeckhauser, R., Swanson, J., and Lockwood, K. 2006. The value of reputation on eBay: A controlled experiment. *Experimental Economics*, 9(2), 79-101.

Shea, C. June 25, 2006. The hidden economy. *The Boston Globe*.

Spiegler, R. 2006. Competition over agents with boundedly rational expectations. *Theoretical Economics*, 1(2): 207-231.

Telser, L.G. 1980. A theory of self-enforcing agreements. *Journal of Business*, 53(1): 27-44.

Xia, L., and K. Monroe. 2004. Price partitioning on the internet. *Journal of Interactive Marketing*, 18.4: 63-73.

Xia, L., K. Monroe, and J. Cox. 2004. The price is unfair! A conceptual framework of price fairness perceptions. *Journal of Marketing*, 68.4: 1-15.

## 6 Tables and Figures

### 6.1 Tables

Table 1: Summary Statistics for Key Firm-Level Variables

| Variable              | Obs    | Mean  | St. Dev. | Min | Max   |
|-----------------------|--------|-------|----------|-----|-------|
| <i>avg_rating</i>     | 35,075 | 3.857 | 0.666    | 1   | 5     |
| <i>resort_fee_sep</i> | 35,075 | 0.042 | 0.200    | 0   | 1     |
| <i>resort_fee_inc</i> | 35,075 | 0.023 | 0.151    | 0   | 1     |
| <i>stars</i>          | 35,075 | 2.576 | 0.619    | 1   | 5     |
| <i>price</i>          | 35,075 | 110.6 | 64.01    | 10  | 1,399 |
| <i>log_price</i>      | 35,075 | 4.595 | 0.451    | 2.3 | 7.2   |
| <i>rooms</i>          | 35,075 | 114.8 | 155.1    | 1   | 5,000 |
| <i>log_rooms</i>      | 35,075 | 4.393 | 0.840    | 0   | 8.5   |
| <i>chain_affil</i>    | 35,075 | 0.651 | 0.477    | 0   | 1     |
| <i>optional_fees</i>  | 35,075 | 0.266 | 0.529    | 0   | 4     |

Note: The full sample of hotels listed on Expedia/Hotels.com as of July 2015 consisted of 46,329 properties. For this analysis, I include only those hotels with complete data for each of the above variables. The most commonly missing datapoint was price.

Table 2: Correlation Matrix for Key Firm-Level Variables

|                | avg_rating | resort_fee_sep | resort_fee_inc | stars   | log_price | log_rooms | chain_affil | optional_fees |
|----------------|------------|----------------|----------------|---------|-----------|-----------|-------------|---------------|
| avg_rating     | 1.000      |                |                |         |           |           |             |               |
| resort_fee_sep | 0.026*     | 1.000          |                |         |           |           |             |               |
| resort_fee_inc | 0.008      | -0.032*        | 1.000          |         |           |           |             |               |
| stars          | 0.548*     | 0.215*         | 0.071*         | 1.000   |           |           |             |               |
| log_price      | 0.561*     | 0.134*         | 0.080*         | 0.635*  | 1.000     |           |             |               |
| log_rooms      | 0.081*     | 0.135*         | -0.005         | 0.362*  | 0.105*    | 1.000     |             |               |
| chain_affil    | 0.177*     | -0.166*        | -0.120*        | -0.066* | -0.092*   | 0.296*    | 1.000       |               |
| optional_fees  | -0.013     | 0.056*         | 0.065*         | 0.229*  | 0.120*    | 0.225*    | -0.047*     | 1.000         |

\*  $p < 0.01$

Table 3: Cross-Sectional OLS Results

|                                     | Dependent Variable: Average (Cumulative) Hotel Rating |                    |                    |                    |                    |                    |
|-------------------------------------|-------------------------------------------------------|--------------------|--------------------|--------------------|--------------------|--------------------|
|                                     | (1)                                                   | (2)                | (3)                | (4)                | (5)                | (6)                |
| resort_fee_sep                      | 0.09***<br>(0.03)                                     | -0.04*<br>(0.02)   | -0.05***<br>(0.02) | -0.06***<br>(0.02) | -0.05***<br>(0.01) | -0.05***<br>(0.02) |
| resort_fee_inc                      | 0.04*<br>(0.02)                                       | 0.02<br>(0.02)     | -0.01<br>(0.02)    | -0.03*<br>(0.02)   | -0.03*<br>(0.02)   | -0.02<br>(0.02)    |
| stars                               |                                                       | 0.63***<br>(0.01)  | 0.49***<br>(0.01)  | 0.47***<br>(0.01)  | 0.43***<br>(0.01)  | 0.39***<br>(0.01)  |
| log_rooms                           |                                                       | -0.17***<br>(0.01) | -0.15***<br>(0.01) | -0.18***<br>(0.01) | -0.16***<br>(0.01) | -0.12***<br>(0.01) |
| log_price                           |                                                       |                    | 0.32***<br>(0.01)  | 0.32***<br>(0.01)  | 0.48***<br>(0.01)  | 0.54***<br>(0.02)  |
| optional_fees                       |                                                       |                    | -0.04***<br>(0.01) | -0.03***<br>(0.01) | -0.02***<br>(0.00) | -0.01<br>(0.01)    |
| Chain Fixed Effects <sup>a</sup>    | -                                                     | Yes                | Yes                | Yes                | Yes                | Yes                |
| Detailed Firm Controls <sup>a</sup> | -                                                     | -                  | -                  | Yes                | Yes                | Yes                |
| Market Fixed Effects                | -                                                     | -                  | -                  | -                  | MSA                | ZIP-5              |
| Constant                            | 3.85***<br>(0.01)                                     | 2.75***<br>(0.03)  | 1.52***<br>(0.05)  | 1.66***<br>(0.05)  | 0.98***<br>(0.06)  | 0.64***<br>(0.09)  |
| Observations                        | 35,075                                                | 35,075             | 35,075             | 35,075             | 35,075             | 35,075             |
| R-squared                           | 0.00                                                  | 0.56               | 0.59               | 0.60               | 0.64               | 0.75               |

Standard errors in parentheses reflect clustering at the ZIP-5 level

(8,271 clusters)

\*\*\*  $p < 0.01$ , \*\*  $p < 0.05$ , \*  $p < 0.1$ <sup>a</sup>See Tables A2 and A3 in the Appendix for details.

Table 4: Resort Fee Policy Changes (April 2015 - March 2016)

|                | Resort Fee Is: |            | Total Hotels |
|----------------|----------------|------------|--------------|
|                | Separate       | Included   |              |
| <b>Adopted</b> | 182            | 104        | <b>286</b>   |
| <b>Dropped</b> | 66             | 67         | <b>133</b>   |
| <b>Total</b>   | <b>248</b>     | <b>171</b> | <b>419</b>   |

*Figures reflect only those policy changes that are eligible for inclusion in the Section 4 difference-in-differences analysis (i.e., those where at least one rating is available both pre- and post- policy change)*

Table 5: Summary Statistics for Key Review-Level Variables

|                                | Overall | Expedia | TripAdvisor | <i>Difference<br/>In Means</i> |
|--------------------------------|---------|---------|-------------|--------------------------------|
| <b>Observations</b>            | 120,498 | 65,040  | 55,458      |                                |
| “Included” Fee Hotels          | 42,876  | 23,135  | 19,741      |                                |
| “Separate” Fee Hotels          | 77,622  | 41,905  | 35,717      |                                |
| <b>Average Rating</b>          | 3.98    | 3.87    | 4.11        | −0.24*                         |
| “Included” Fee Hotels          | 3.95    | 3.84    | 4.09        | −0.25*                         |
| “Separate” Fee Hotels          | 3.99    | 3.89    | 4.12        | −0.23*                         |
| <b>Rating Distribution:</b>    |         |         |             |                                |
| <i>(“Included” Fee Hotels)</i> |         |         |             |                                |
| 1                              | 0.053   | 0.053   | 0.052       | 0.002                          |
| 2                              | 0.072   | 0.082   | 0.061       | 0.021*                         |
| 3                              | 0.154   | 0.182   | 0.122       | 0.061*                         |
| 4                              | 0.310   | 0.340   | 0.275       | 0.065*                         |
| 5                              | 0.411   | 0.343   | 0.491       | −0.148*                        |
| <b>Trip Type:</b>              |         |         |             |                                |
| <i>(“Included” Fee Hotels)</i> |         |         |             |                                |
| Business <i>(of specified)</i> | 0.149   | 0.149   | 0.148       | 0.001                          |
| Couple <i>(of specified)</i>   | 0.346   | 0.348   | 0.344       | 0.004                          |
| Family <i>(of specified)</i>   | 0.400   | 0.396   | 0.402       | −0.006                         |
| Friends <i>(of specified)</i>  | 0.106   | 0.107   | 0.106       | 0.001                          |

Stars indicate significance at the 1% level for a t-test with unequal variance between groups.

Table 6: Difference-In-Differences Estimation Results

|                                  | Dependent Variable: Traveler Rating |                     |                     |                     |                     |
|----------------------------------|-------------------------------------|---------------------|---------------------|---------------------|---------------------|
|                                  | (1)                                 | (2)                 | (3)                 | (4)                 | (5)                 |
| tripadv * resort_fee             | -0.15 ***<br>(0.05)                 | -0.14 ***<br>(0.03) | -0.14 ***<br>(0.03) | -0.13 ***<br>(0.03) | -0.13 ***<br>(0.03) |
| tripadv                          | 0.32 ***<br>(0.05)                  | 0.11 ***<br>(0.03)  | 0.10 ***<br>(0.03)  | 0.10 ***<br>(0.03)  | 0.08 **<br>(0.03)   |
| resort_fee                       | 0.12 **<br>(0.06)                   | 0.07 ***<br>(0.02)  | 0.06 **<br>(0.02)   | 0.05 **<br>(0.02)   | 0.05 **<br>(0.02)   |
| business_trav                    |                                     |                     |                     |                     | 0.05*<br>(0.03)     |
| couple_trav                      |                                     |                     |                     |                     | 0.10 ***<br>(0.02)  |
| family_trav                      |                                     |                     |                     |                     | 0.05 ***<br>(0.02)  |
| friends_trav                     |                                     |                     |                     |                     | 0.15 ***<br>(0.02)  |
| Firm Fixed Effects               | -                                   | Yes                 | Yes                 | Yes                 | Yes                 |
| Month Fixed Effects              | -                                   | -                   | Yes                 | Yes                 | Yes                 |
| Chain Fixed Effects <sup>a</sup> | -                                   | -                   | -                   | Yes                 | Yes                 |
| Firm Controls <sup>b</sup>       | -                                   | -                   | -                   | Yes                 | Yes                 |
| Constant                         | 3.78 ***<br>(0.06)                  | 3.90 ***<br>(0.02)  | 3.89 ***<br>(0.03)  | 3.63 ***<br>(0.32)  | 3.63 ***<br>(0.32)  |
| Observations                     | 42,876                              | 42,876              | 42,876              | 42,082              | 42,082              |
| R-squared                        | 0.01                                | 0.21                | 0.21                | 0.21                | 0.22                |

Standard errors in parentheses reflect clustering at the firm level

(171 clusters in Models 1-3 and 169 clusters in Models 4-5)

\*\*\*  $p < 0.01$ , \*\*  $p < 0.05$ , \*  $p < 0.1$ <sup>a</sup> These are allowed to vary over time. See Table A3 in the Appendix for details on chain fixed effects.<sup>b</sup> These are allowed to vary over time, and include **stars**, **log-price**, **optional\_fees**, and all of the detailed controls listed in Table A2.

Table 7: Hotel Characteristics By Resort Fee Regime

|                                | Mean During Period:   |                    | Change | T-Test Significant<br>at 5% Level? |
|--------------------------------|-----------------------|--------------------|--------|------------------------------------|
|                                | Without<br>Resort Fee | With<br>Resort Fee |        |                                    |
| <i>major_chain</i>             | 32.9%                 | 31.7%              | -      | No                                 |
| <i>optional_fees</i>           | 43.5%                 | 42.4%              | -      | No                                 |
| <b>Amenities:</b>              |                       |                    |        |                                    |
| <i>water_onsite</i>            | 10.4%                 | 12.5%              | +      | No                                 |
| <i>winter_onsite</i>           | 6.8%                  | 7.7%               | +      | No                                 |
| <i>tennis_onsite</i>           | 11.4%                 | 11.0%              | -      | No                                 |
| <i>golf_onsite</i>             | 7.3%                  | 7.7%               | +      | No                                 |
| <i>casino</i>                  | 2.6%                  | 3.2%               | +      | No                                 |
| <i>restaurants</i>             | 0.69                  | 0.71               | +      | No                                 |
| <i>concierge</i>               | 33.4%                 | 40.0%              | +      | <b>Yes</b>                         |
| <i>turndown</i>                | 9.2%                  | 9.4%               | +      | No                                 |
| <i>indoor_pools</i>            | 0.23                  | 0.23               | +      | No                                 |
| <i>outdoor_pools</i>           | 0.71                  | 0.71               | +      | No                                 |
| <i>beach_pool_amenities</i>    | 14.5%                 | 21.1%              | +      | <b>Yes</b>                         |
| <i>fitness_classes</i>         | 4.3%                  | 6.8%               | +      | No                                 |
| <i>Health Club Onsite</i>      | 7.2%                  | 9.8%               | +      | No                                 |
| <i>Full-Service Spa Onsite</i> | 11.6%                 | 14.3%              | +      | No                                 |

See Tables A1 and A2 for details on these variables. The last two items represent specific values for the categorical variables *fitness* and *spa*.

Table 8: Reviews Flagged as “Fake” by Resort Fee Regime

|             | Without<br>Resort Fee         | With<br>Resort Fee            | Difference | T-Test Significant<br>at 5% Level? |
|-------------|-------------------------------|-------------------------------|------------|------------------------------------|
| TripAdvisor | 12.7%<br><i>19 out of 150</i> | 13.3%<br><i>20 out of 150</i> | 0.7%       | No                                 |
| Expedia     | 8.0%<br><i>12 out of 150</i>  | 6.7%<br><i>10 out of 150</i>  | -1.3%      | No                                 |

A random sample of 150 reviews in each of these four categories was selected for manual testing.

“Fake” reviews are those flagged as deceptive (vs. truthful) by *reviewskeptc.com*, accessed September 2016 (see citation for Ott et al).



Table 9: Difference-in-Differences Results for Various Hotel Sub-Categories

| Hotel Sub-Sample:                                       | Coefficient Estimates For:               |                                                                | Number of Reviews     |
|---------------------------------------------------------|------------------------------------------|----------------------------------------------------------------|-----------------------|
|                                                         | Main Effect<br>( <i>resort_fee_inc</i> ) | Treatment Effect<br>( <i>tripadv</i> * <i>resort_fee_inc</i> ) |                       |
| Low-Tier<br>( <i>stars</i> ≤ 2)                         | −0.03<br>(0.07)                          | −0.34 * **<br>(0.09)                                           | 6,217<br>(44 Hotels)  |
| Mid-Tier<br>(2 < <i>stars</i> < 3.5)                    | 0.05<br>(0.03)                           | −0.11 * *<br>(0.05)                                            | 21,493<br>(84 Hotels) |
| High-Tier<br>( <i>stars</i> ≥ 3.5)                      | 0.06<br>(0.04)                           | −0.08<br>(0.06)                                                | 14,372<br>(41 Hotels) |
| <i>Within the Low-Tier Segment (stars ≤ 2):</i>         |                                          |                                                                |                       |
| Low-Price <sup>a</sup>                                  | 0.01<br>(0.07)                           | −0.35 * *<br>(0.13)                                            | 3,090<br>(20 Hotels)  |
| High-Price <sup>a</sup>                                 | −0.10<br>(0.12)                          | −0.35 * **<br>(0.10)                                           | 3,127<br>(24 Hotels)  |
| Resort Amenities <i>Not</i> Provided <sup>b</sup>       | −0.03<br>(0.07)                          | −0.31 * **<br>(0.09)                                           | 5,630<br>(41 Hotels)  |
| Resort Amenities Provided <sup>b</sup>                  | −0.36<br>(0.81)                          | −0.31<br>(0.20)                                                | 587<br>(3 Hotels)     |
| <i>Within the Mid/High-Tier Segment (stars &gt; 2):</i> |                                          |                                                                |                       |
| Low-Price <sup>a</sup>                                  | 0.02<br>(0.04)                           | −0.11 * *<br>(0.05)                                            | 20,982<br>(66 Hotels) |
| High-Price <sup>a</sup>                                 | 0.13 * **<br>(0.04)                      | −0.12 * *<br>(0.05)                                            | 14,883<br>(59 Hotels) |
| Resort Amenities <i>Not</i> Provided <sup>b</sup>       | 0.04<br>(0.03)                           | −0.12 * **<br>(0.05)                                           | 21,872<br>(77 Hotels) |
| Resort Amenities Provided <sup>b</sup>                  | 0.05<br>(0.03)                           | −0.06<br>(0.06)                                                | 13,993<br>(48 Hotels) |

Standard errors parentheses reflect clustering at the firm level

\*\*\*  $p < 0.01$ , \*\*  $p < 0.05$ , \*  $p < 0.1$ *Estimates reflect sub-sample results for the Model (5) specification in Table 6.*<sup>a</sup> Low-price (high-price) defined as hotels with price less than or equal to (greater than) the corresponding MSA-market average for hotels of the same star category.<sup>b</sup> Resort amenities are defined to include: beach and/or pool amenities (e.g., cabanas, towel service, etc.), full-service spas, health clubs, and on-site golfing. If a hotel offers any of these, it is coded as providing resort amenities.

## 6.2 Figures

Figure 1: Cross-Sectional Rating Densities for Hotels With vs. Without Resort Fees

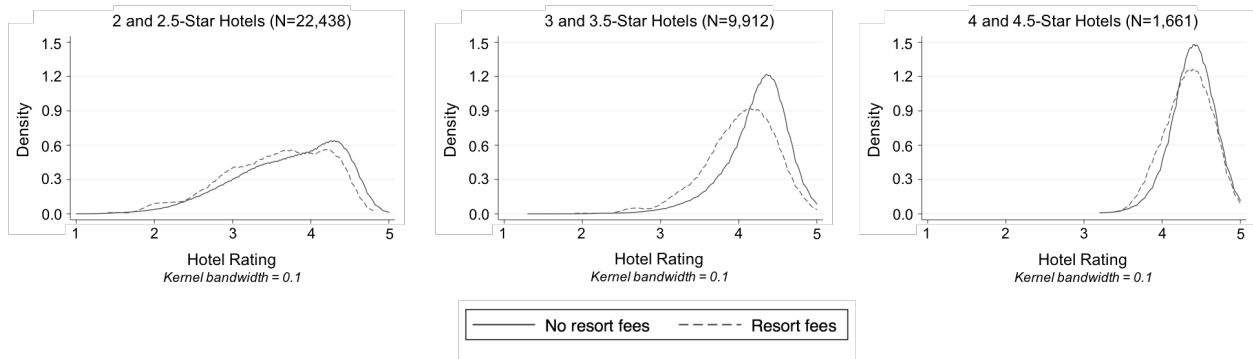


Figure 2: “Included” vs. “Separate” Resort Fees on Hotels.com

**\$209**  
nightly price  
Excludes daily resort fee

← Price displayed prior to final booking page

**\$295**  
nightly price  
Excludes \$22 daily resort fee

← Price displayed prior to final booking page

### Your Reservation Summary



**South Seas Hotel**  
Miami Beach, FL, US  
866-925-8705

**Good 3.5 / 5**  
Hotels.com Guest reviews

  
TripAdvisor Traveler Rating

Check in: Friday, September 18, 2015  
Check out: Saturday, September 19, 2015  
**1 night stay**

 **Includes special offer**  
Free Breakfast Included Daily

**Classic Room, 2 Double Beds** Continental Breakfast, Free Wireless Internet

|                            |                 |
|----------------------------|-----------------|
| Friday, September 18, 2015 | \$209.34        |
| Taxes and fees             | \$43.04         |
| <b>Total to pay now</b>    | <b>\$252.38</b> |

(including taxes and fees) ⓘ

Full payment will be charged when you book this hotel.

### Your Reservation Summary



**Hotel Breakwater South Beach**  
Miami Beach, FL, US  
866-925-8705

**Good 3.8 / 5**  
Hotels.com Guest reviews

  
TripAdvisor Traveler Rating

Check in: Friday, September 18, 2015  
Check out: Saturday, September 19, 2015  
**1 night stay**

 **Includes special offer**  
Save 20%

**Standard Room, 1 King Bed, Courtyard View, C...** Full Breakfast

|                            |                 |
|----------------------------|-----------------|
| Friday, September 18, 2015 | \$295.45        |
| Taxes and fees             | \$38.40         |
| <b>Total to pay now</b>    | <b>\$333.85</b> |

(including taxes and fees) ⓘ

Payment for the room will be taken when you book this hotel

**The hotel will charge an additional fee at check in or check out ⓘ** **\$22.00**

**Total cost for this booking** **\$355.85**

*The South Seas Hotel and the Hotel Breakwater – both 3.5 star hotels in Miami – collect payment for resort fees differently on the Expedia platform. On the left, the total price **includes** a resort fee of \$15 (embedded within the “taxes and fees” line). On the right, total price does **not** include a resort fee of \$22.00, which is instead collected at the property (as noted in fine print at the bottom of the page).*

Figure 3: Resort Fees (“Included”) on Hotels.com vs. Elsewhere



**Fairmont Scottsdale Princess**  
Scottsdale, AZ, US  
866-925-8705

**Excellent 4.5 / 5**  
Hotels.com Guest reviews

  
TripAdvisor Traveler Rating

Check in                      Tuesday, November 22, 2016  
Check out                    Wednesday, November 23, 2016  
1 night stay

---

Fairmont Room, 1 King Bed, Non Smoking  
Tuesday, November 22, 2016                      \$299.49

Taxes and fees                                              \$75.87

---

Total to pay now                      **\$375.36**  
including taxes and fees ⓘ

Full payment will be charged when you book this hotel.



**RESERVATION DETAILS**

**ARRIVAL DATE**  
22-Nov-2016

**DEPARTURE DATE**  
23-Nov-2016

**TOTAL NIGHTS**  
1

**ROOM RATE**  
\$299 USD

[View Room](#)  
[Rate Summary](#)

---

**TOTAL:**  
**\$ 340<sup>62</sup>**

Includes 7.27% Rm Tx 5% Rm Occ Tx  
1.65% Rm City Tx . Does not include  
30 USD Resort Fee plus taxes.

*The total price for the Fairmont Scottsdale is displayed differently depending on where a traveler books. On Hotels.com, the large-font total **includes** the resort fee, while the Fairmont’s own website only displays this additional charge in fine print.*

Figure 4: Traveler Ratings Before and After Resort Fee Adoption

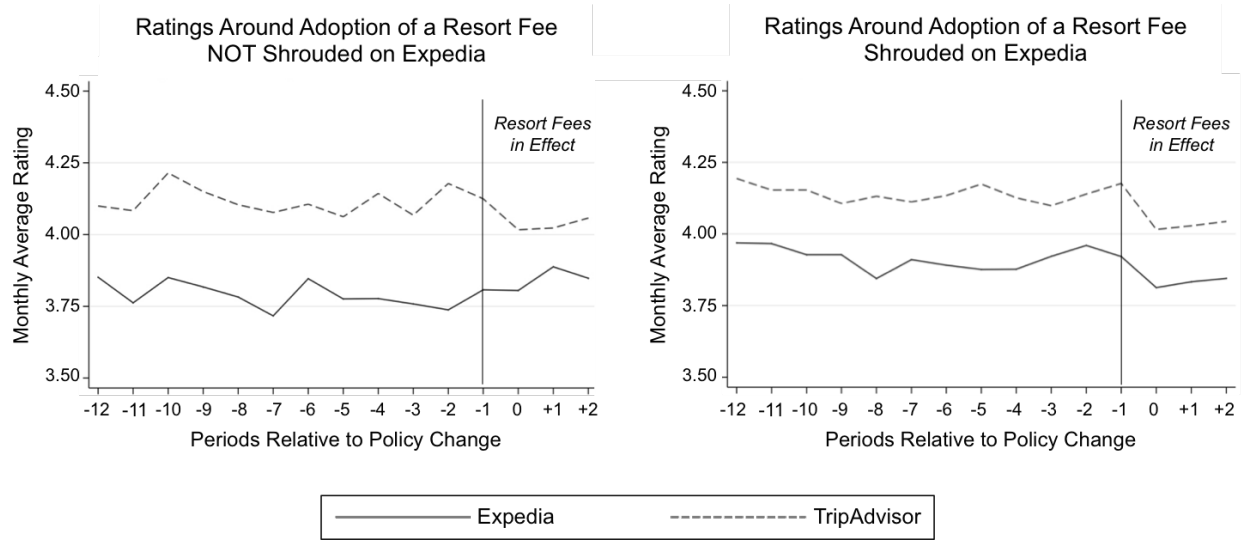
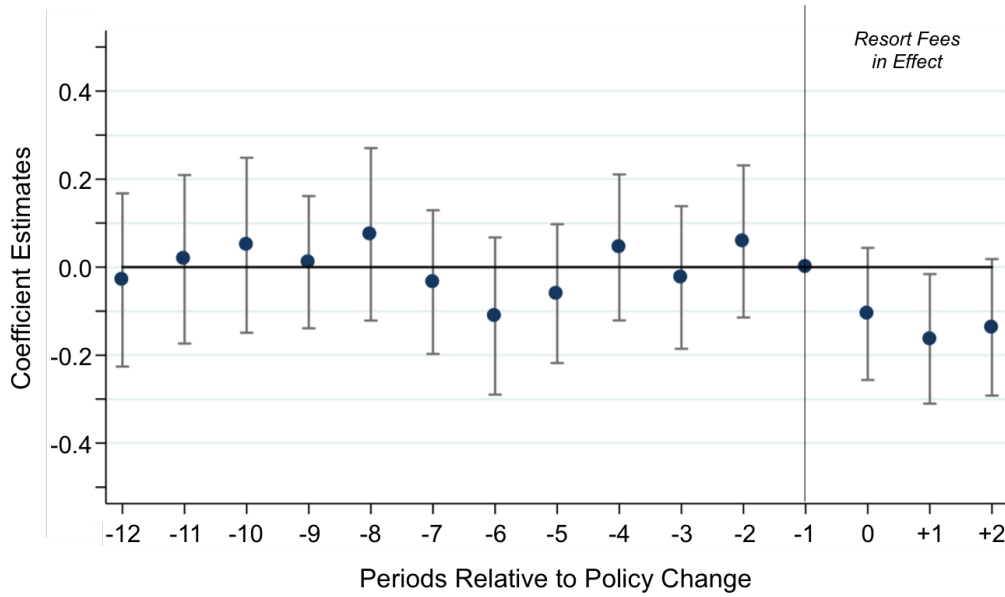


Figure 5: Pre-Treatment Trends Around Resort Fee Adoption



Review data in this specification corresponds to that in the left-hand side of Figure 4, which includes hotels that adopted resort fees **not** shrouded on Expedia. The dependent variable is individual traveler rating; circles indicate coefficient estimates on interactions between time-to-treatment dummy variables and an indicator variable for the “treatment group” (TripAdvisor). Bars represent 95% confidence intervals. The last month before resort fee adoption is the omitted interaction.

## Appendix: Additional Tables and Figures

Table A1: Key Variable Descriptions: Cross-Sectional Expedia Data

| <i>Variable</i>        | <i>Description</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <i>avg_rating</i>      | The average cumulative traveler rating for a hotel, on a scale of 1 to 5 (displayed in increments of 0.1 points). Average ratings reflect traveler reviews written on Hotels.com and Expedia.com beginning in 2012. These reviews are submitted <i>only</i> by individuals who booked through an Expedia platform.                                                                                                                                                                                                               |
| <i>resort_fee_sep:</i> | A dummy variable, equal to 1 if the hotel charges a resort fee that is <i>not</i> included in the total price paid up front by Hotels.com/Expedia customers.                                                                                                                                                                                                                                                                                                                                                                     |
| <i>resort_fee_inc:</i> | A dummy variable, equal to 1 if the hotel charges a resort fee that <i>is</i> included in the total price paid up front by Hotels.com/Expedia customers.                                                                                                                                                                                                                                                                                                                                                                         |
| <i>stars:</i>          | The star category assigned to each individual hotel by Expedia. Properties can receive scores between 1 and 5 in half-star increments, which reflect the level of accommodations and services that the hotel offers. It is important to distinguish between a hotel's star categorization and its average traveler rating; while higher-star hotels do tend to receive higher traveler ratings on average, it is possible for a 2-star hotel to have very high traveler ratings, while a 4-star hotel might receive low ratings. |
| <i>chain_affil:</i>    | A dummy variable equal to 1 if the property is affiliated with a major hotel chain (see Table A3 for a list of chains included in the calculation of this variable).                                                                                                                                                                                                                                                                                                                                                             |
| <i>log_price:</i>      | The natural log of the lowest nightly rate displayed on each hotel's main Hotels.com page. This nightly price may correspond to any date within two weeks of the date on which the page was referenced. In all cases, this price reflects only the base rate (and does <i>not</i> include the amount of the resort fee).                                                                                                                                                                                                         |
| <i>log_rooms:</i>      | The natural log of the total number of rooms at a given hotel as listed by Hotels.com. <sup>31</sup>                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <i>optional_fees:</i>  | Dummy variables indicate whether or not a hotel charges for the following optional add-ons: wireless internet in the guest rooms ( <i>wifi_fee</i> ), self-parking ( <i>selfpark_fee</i> ), late check-out ( <i>latechk_fee</i> ), and infant cribs ( <i>crib_fee</i> ). The variable <i>optional_fees</i> is the sum of these dummy variables. It reflects the number of optional add-on fees a hotel charges out of these four major categories.                                                                               |

Table A2: Descriptions for Detailed Controls: Cross-Sectional Expedia Data

| <i>Variable(s)</i>                 | <i>Description</i>                                                                                                                                                                                                                                                   |
|------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <i>water_onsite, winter_onsite</i> | Dummy variables indicating whether or not a hotel offers (respectively) water activities (e.g., surfing, sailing, snorkeling, etc.) and winter activities (e.g., skiing, sledding, etc.) on site.                                                                    |
| <i>tennis_onsite, golf_onsite</i>  | Dummy variables indicating whether or not a hotel offers (respectively) tennis and golf on site.                                                                                                                                                                     |
| <i>casino</i>                      | Dummy variable indicating whether or not a hotel has a casino on site.                                                                                                                                                                                               |
| <i>fitness</i>                     | Categorical variable indicating the type of fitness facilities a hotel property offers. Possible values include: “None,” “24-Hour Fitness Facilities On Site,” “Fitness Facilities (Hours Vary) On Site,” “Health Club On Site,” and “Use of Nearby Fitness Center.” |
| <i>fitness_classes</i>             | Dummy variable indicating whether or not a hotel offers fitness classes.                                                                                                                                                                                             |
| <i>spa</i>                         | Categorical variable indicating spa services offered by a hotel property. Possible values include: “None,” “Full-Service Spa On Site,” and “Spa Services / Spa Treatment Rooms On Site.”                                                                             |
| <i>restaurants</i>                 | The number of restaurants that a hotel property offers on site.                                                                                                                                                                                                      |
| <i>roomserv</i>                    | Categorical variable indicating the type of room service a hotel property offers. Possible values include: “None,” “Room Service (Limited Hours),” and “24-Hour Room Service.”                                                                                       |
| <i>concierge</i>                   | Dummy variable indicating whether or not a hotel offers concierge service.                                                                                                                                                                                           |
| <i>turndown</i>                    | Dummy variable indicating whether or not a hotel offers turndown service.                                                                                                                                                                                            |
| <i>indoor_pools, outdoor_pools</i> | The number of (respectively) indoor and outdoor pools a hotel offers.                                                                                                                                                                                                |
| <i>beach_pool_amenities</i>        | Dummy variable indicating whether or not a hotel offers beach and/or pool services (e.g., loungers, towels, cabanas, etc.).                                                                                                                                          |

Table A3: Hotel Chains

The variable *chain\_affil* is equal to 1 if a hotel is affiliated with any of these chains. This list also corresponds to the set of chains included as chain-level fixed effects in regression analysis.

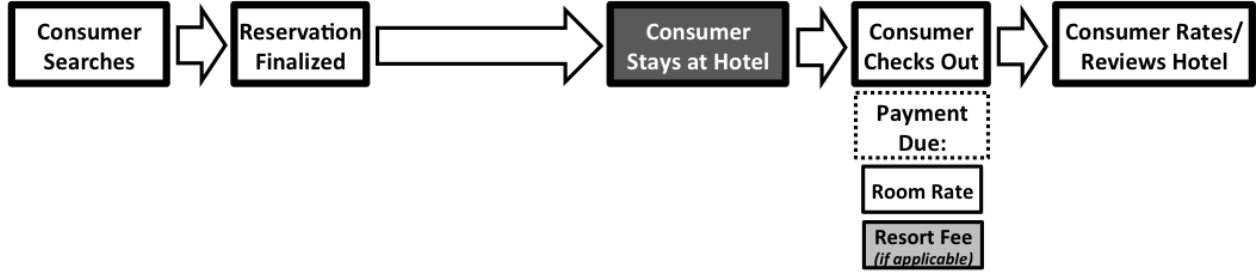
| <i>Group</i>                          | <i>Chains Included</i>                                                                                                                                                                                                                       |
|---------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Best Western Hotel Group:</b>      | Best Western, Best Western Plus, Best Western Premier                                                                                                                                                                                        |
| <b>Carlson Rezidor Hotel Group:</b>   | Country Inn & Suites, Park Inn by Radisson, Park Plaza, Radisson, Radisson Blu                                                                                                                                                               |
| <b>Choice Hotels:</b>                 | Ascend, Cambria, Clarion, Comfort Inn (& Suites), Econolodge, Mainstay, Quality Inn (& Suites), Rodeway, Sleep Inn, Suburban Extended Stay                                                                                                   |
| <b>Hilton Worldwide:</b>              | Conrad, DoubleTree, Embassy Suites, Hampton Inn (& Suites), Hilton, Hilton Garden Inn, Home2, Homewood, Waldorf Astoria                                                                                                                      |
| <b>Hyatt Hotels Corp:</b>             | Andaz, Grand Hyatt, Hyatt, Hyatt House, Hyatt Place, Hyatt Regency, Park Hyatt                                                                                                                                                               |
| <b>InterContinental Hotels Group:</b> | Candlewood, Crowne Plaza, EVEN Hotel, Holiday Inn, Holiday Inn Express, Hotel Indigo, InterContinental, Kimpton, Staybridge                                                                                                                  |
| <b>La Quinta:</b>                     | La Quinta                                                                                                                                                                                                                                    |
| <b>Marriott International:</b>        | Autograph Collection, Courtyard by Marriott, Fairfield, Gaylord, JW Marriott, Marriott, Renaissance, Residence, Ritz-Carlton, Springhill, Towneplace                                                                                         |
| <b>Motel 6:</b>                       | Motel 6, Studio 6                                                                                                                                                                                                                            |
| <b>Starwood Hotels and Resorts:</b>   | Aloft, Element, Four Points, Le Meridien, Luxury Collection, Sheraton, St. Regis, W Hotels, Westin                                                                                                                                           |
| <b>Wyndham Worldwide:</b>             | Baymont Inn (& Suites), Days Inn (& Suites), Hawthorn Suites, Howard Johnson, Howard Johnson Express, Knights Inn, Microtel, Planet Hollywood, Ramada, Super 8, Travelodge, Tryp, Wingate by Wyndham, Wyndham, Wyndham Garden, Wyndham Grand |



Table A4: Descriptions for Key Review-Level Variables

| <i>Variable</i>  | <i>Description</i>                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <i>rating</i>    | Individual traveler rating for a hotel, provided as an integer value between 1 and 5. (It is important to distinguish between <i>rating</i> and <i>average_rating</i> , the variable utilized in the preceding cross-sectional analysis.) Recall that all ratings on Expedia sites are submitted <i>only</i> by travelers who book on Expedia, whereas TripAdvisor ratings may be submitted by anyone, regardless of how they have booked their room. |
| <i>tripadv</i>   | A dummy variable, equal to 1 if the review/rating comes from TripAdvisor, and equal to 0 if the review/rating comes from Expedia. In the subsequent difference-in-differences analysis, this effectively serves as an indicator for whether or not a review comes from the “treatment group.”                                                                                                                                                         |
| <i>trip_type</i> | A categorical variable indicating the purpose/type of the reviewer’s trip. Possible values include: “Business,” “Family,” “Friends,” “Couple,” and “Other/Unspecified.”                                                                                                                                                                                                                                                                               |

Figure A1: Timeline for a Typical (Non-Expedia) Hotel Transaction



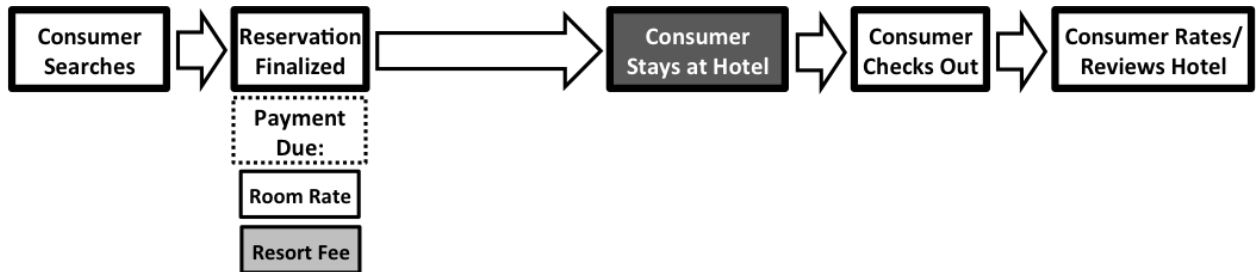
For travelers booking rooms via channels other than Expedia, payment is typically not due until the traveler departs the hotel. Because resort fees are often not clearly disclosed at the time of booking, travelers may not notice them (or not fully account for them) until payment is actually requested at the end of their stay.

Figure A2: Timelines for a Typical Expedia Hotel Transaction

(a) Resort Fees Charged Separately ( $resort\_fee\_sep=1$ )



(b) Resort Fees Included Up Front ( $resort\_fee\_inc=1$ )



On Expedia, hotels that charge resort fees can choose to either include this fee in the total price charged up front (b) or to charge the fee separately to the traveler at the hotel (a). In the case of (a), consumers may not notice (or fully account for) the fee at the time of booking, in parallel with the situation depicted in Figure A1 above. In (b), however, there is no opportunity for consumers to be surprised by the fee, as it is (by definition) fully disclosed when its payment is required at the time of booking.